Non-Equilibrium Properties Reveal the Limits of Specialist and Foundation Machine-Learned Force Fields

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Abstract

Machine-learned force fields (MLFFs), particularly those arising from pre-trained foundation models, offer an enticing compromise between *ab initio* accuracy and the larger scope possible with semi-empirical force field-derived molecular dynamics with respect to length and time scales. Despite their promise, a fundamental question remains: To what extent should one rely on specialist models, fine-tuned foundation models, or a hybrid approach? Trade-offs in data efficiency, robustness, and non-equilibrium fidelity remain poorly understood, as does understanding how different training paradigms encode physical information. To assist with this decision, we introduce a unified benchmarking framework based on non-equilibrium configurational probes that span interpolation and extrapolation regimes. Migration pathways computed via Nudged Elastic Band calculations serve as the primary diagnostic probe, complemented by equilibrium and mechanical tests. Using Cr-doped Sb_2Te_3 as a representative 2D material for which non-equilibrium properties are critically important, we evaluate multiple training strategies within the MACE architecture. While all models reproduce equilibrium structures, they diverge sharply in their ability to adequately capture non-equilibrium processes. Fine-tuned MACE models improve selected kinetic probes but can degrade long-range mechanical response. "Zero-shot" MACE foundation models exhibit overly smoothed potentials and specialist models can fail for high-energy transitions. Latent-space and surrogate-error diagnostics are consistent with distinct model representations across training paradigms, supporting the conclusion that equilibrium metrics alone are insufficient.

This work demonstrates migration-based non-equilibrium probes as a data-efficient candidate framework for MLFF evaluation, developed and tested on a representative material system for which such probes are important, and provides practical guidance for robust MLFF development.

Keywords: Machine-learned force fields (MLFFs), foundation models, fine-tuning, defect migration pathways, 2D materials

1 Introduction

Machine-Learned Force Fields (MLFFs) [1–14] have emerged as a transformative solution to the dilemma of accuracy-versus-cost in molecular simulation. The development of large-scale, pre-trained so-called "foundation models," such as CHGNet, [15] MEGNet, [16] M3GNet, [12] MACE-MP, [4] MatterSim, [17] and UMA, [18] marks a paradigm shift to try to resolve this dilemma. [19]. However, this shift presents researchers with a critical question: For a specific system, is it more effective to invest significant resources to build a robust *specialist* model from scratch, [20–22] or to adapt a *generalist* foundation model through fine-tuning? While generic benchmarks like Matbench, [23] JARVIS-Leaderboard, [24] and LeMat-GenBench, [25] validate their broad utility to predict equilibrium properties for which they were trained, their reliability for specific, high-fidelity applications remains an open question. [26–28] A valid concern is that these models fail to capture potentially more nuanced interactions that govern critical processes that shape the materials and their processing. [29–32] Fine-tuning is a promising solution, but using naïve strategies can lead to *"catastrophic forgetting"*, where general knowledge about the system is erased. Key questions of data efficiency, [33, 34] stability, [35, 36] and transferability [37, 38] for non-equilibrium processes remain largely unsolved. [39–41]

A rigorous benchmark must specify both the configuration distribution and the observables being measured. Conventional MLFF validation typically evaluates pointwise energy and force errors near equilibrium; migration and collective-displacement pathways test the same potential energy surface away from equilibrium. We therefore use transition-state and strained-lattice configurations as concrete process tests, and separate equilibrium pointwise accuracy, fixed-path energetic fidelity, and self-relaxed path stability in Sec. 2.3.

To demonstrate and validate this benchmarking framework, we employ doped two-dimensional (2D) van der Waals (vdW) materials as an ideal test bed. The discovery and design of novel 2D vdW materials [42–46] continues to drive innovation in spintronics, [47] quantum devices, [48–50] and energy technologies. [51, 52] A particularly powerful strategy for tuning the properties of these layered materials involves doping, the insertion of "foreign" atoms between vdW layers. [42, 53] This approach has enabled remarkable advances in materials design, ranging from lithium-ion batteries [54, 55] to the engineering of magnetic semiconductors [56] and quantum anomalous Hall systems. [57] Crucially, their layered structure offers distinct environments for

atomic motion, making them an excellent probe for testing MLFF capabilities across different chemical contexts within a single system.

Among 2D materials, topological insulators such as antimony telluride (Sb_2Te_3) [58, 59] represent an especially promising class for doping studies. The doping of transition metals has proven to be effective across multiple topological systems: Iron in Bi_2Se_3 creates ferromagnetic order [60–62]. Manganese in Bi_2Te_3 enables tunable magnetic properties. [63, 64] And vanadium in Sb_2Te_3 modifies its electronic structure. [65, 66] The doping of chromium (Cr) into Sb_2Te_3, the focus of this work, offers a compelling route to engineer its magnetic and topological properties for spintronic applications. [67–69] Predicting the stability and dynamics of these complex systems is essential for guiding experimental synthesis, a task for which large-scale Molecular Dynamics (MD) simulations are extremely powerful [70, 71] provided that a suitably representative force field is available. However, the atomic-scale processes governing functionality, including dopant migration and thermal stability, occur on time- and length- scales far beyond the reach of first-principles methods like density functional theory (DFT). [72–75]

This work addresses these gaps by presenting a systematic benchmark of bespoke training and fine-tuning strategies for the case study of Cr-doped Sb_2Te_3 using the MACE architecture (Fig. 1). By combining latent-space analysis with migration-based probes, we move beyond traditional accuracy metrics to evaluate the dynamic stability and kinetic predictions of each model. Our study aims to serve as both a practical guide for selecting MLFF training strategies and a diagnostic framework for designing more data-efficient learning loops.

Three recent studies provide complementary benchmarks but address different questions. Alghamdi et al. [76] compare MACE-MPA-0 fine-tuning strategies for Li diffusion in LiF using nanosecond-scale molecular dynamics and Arrhenius transport observables, showing that a few hundred targeted configurations can reproduce a DeePMD reference. Bheemaguli et al. [77] evaluate zero-shot foundation MLIPs coupled to NEB over 574 battery-relevant migration paths, and separate barrier accuracy, conductor classification, and local transition-path geometry; this establishes broad screening utility without comparing specialist and fine-tuned training strategies in a chemically controlled system. Deng et al. [78] identify systematic underprediction of energies, forces, and PES curvature in universal MLIPs across out-of-distribution tasks, and show that sparse high-energy fine-tuning can correct part of this bias. We therefore treat PES softening as an established mechanism that motivates our diagnostics, not as a phenomenon first introduced here. Our contribution is to connect these strands in one doped layered material by comparing training strategies with fixed-path energetic fidelity, self-relaxed NEB stability, local migration, collective interlayer sliding, representation diagnostics, and provenance-aware evaluation.

This manuscript substantially extends our earlier non-archival AI4Mat workshop report [79]; it is not a duplicate publication. The earlier report introduced migration as a probe and provided an initial four-strategy MACE case study. The present journal manuscript adds a formal separation of fixed-path scoring from model-in-the-loop NEB optimization, distinct in-gap and higher-distortion migration probes, interlayer sliding, representation and surrogate-error diagnostics, grouped/provenance-aware splitting with nearest-neighbor overlap audits, and targeted multi-seed and data-volume controls.

These additions support a more precise claim: migration pathways are direct tests of model behavior away from equilibrium, and their conclusions must remain indexed to the pathway, training provenance, model initialization, and convergence status.

To this end, we address the following key questions:

1. How do bespoke MLFFs, trained from scratch, compare against large pre-trained foundation models in terms of accuracy, stability, and data efficiency for molecular simulation? What are the performance and stability trade-offs of different fine-tuning strategies?
2. Can non-equilibrium probes generated from methods, such as NEB, provide an efficient benchmark for specialist versus generalist models, and what additional cross-system evidence is needed before such probes can be considered generalizable across materials and architectures?
3. Can latent-space analysis reveal failure signatures to guide data acquisition?

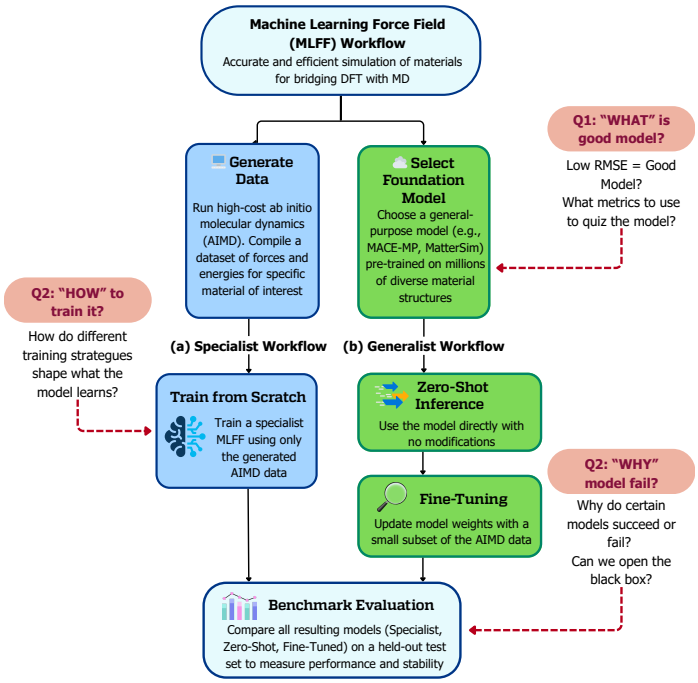


Fig. 1: **Conceptual Workflow Diagram** illustrating two competing workflows: (a) training a specialist MLFF from scratch and (b) fine-tuning a generalist foundation model. The diagram highlights the benchmarking questions addressed in this paper.

2 Methodology

We employed a systematic approach to benchmark various Machine-Learned Force Fields (MLFFs) for Cr-doped Sb_2Te_3[1], comparing specialist models trained from scratch against fine-tuned foundation models.

2.1 Data-Generation and Model-Training

Reference data were generated using density functional theory (DFT) calculations adopting the well-used PBE functional [73] and comprising roughly 20,000 configurations of a $4 \times 2 \times 1$ system of the Sb_2Te_3 crystal [80]. We use a 1 fs timestep for a series of *ab initio* molecular dynamics (AIMD) simulations at three temperatures (300 K, 600 K, 1200 K). We employed the MACE architecture for all models, which is a message-passing neural network for the representation of atomic interactions. [4]

This paper evaluates four distinct training strategies to benchmark their performance, named as follows:

1. **Scratch**: MACE model trained exclusively on our AIMD dataset
2. **Foundation**: Pre-trained MACE-MP model used without fine-tuning
3. **FT-600 K**: Foundation model fine-tuned on 5% of data from 600 K trajectories
4. **FT-MultiT**: Foundation model fine-tuned on multi-temperature data from the three temperatures studied

Revision 1 treats frame-level train/test splitting as insufficient for benchmark claims involving migration pathways. The revised data workflow assigns each entire source trajectory to a single partition (train, validation, or test) rather than shuffling individual frames, so that no trajectory is split across partitions. Under this grouped protocol (1,786/311/267 train/validation/test frames), the nearest-neighbor SOAP cosine distance between any in-gap NEB image and the grouped training set is 5.4×10^{-4}, and the closest comparable Cartesian RMSD is 2.06 Å; both are far outside the exact/near-duplicate thresholds used for this audit (SOAP $\leq 10^{-4}$, RMSD ≤ 0.05 Å), so no NEB-image/training-frame overlap was detected under the declared metrics. The Scratch-5% training subset (89 frames) gives the same result (SOAP 6.4×10^{-4}, RMSD 2.06 Å). Grouped-split retraining across seeds 123/234/345 is complete for FT-600K fine-tuned from the OMAT-FT checkpoint, Scratch, and Scratch-5%, together with a single-seed MACE-MPA-0 fine-tuned comparator (excluded from 3-seed uncertainty estimates); the resulting RMSE and in-gap barrier statistics are reported below and in Table S1.

The reported fine-tuned models are documented as being trained on AIMD frames with real DFT energies and forces, not on NEB-derived exports lacking real force labels. The revised data manifest records source trajectory, split assignment, energy/force key mapping, and checksums for model-training files. The provenance audit covers

[1]This material system serves as an ideal testbed because its anisotropic van der Waals structure provides distinct migration environments: in-gap diffusion within quintuple layers (representing in-distribution scenarios) and interlayer migration across vdW gaps (probing out-of-distribution regimes). This structural duality enables efficient sampling of diverse configurations—from equilibrium states to high-energy transition states—while addressing technologically relevant processes where both thermodynamic stability and kinetic properties are critical for doping engineering.

9,163 frames across the historical and grouped exports; the only nine exact zero-force frames are explicitly labelled one-atom Cr/Sb/Te isolated-atom E_0 references, with zero unclassified and zero NEB-derived zero-force frames. The complete manifest and audit code are in release commit `f5c6d19`.

2.2 Evaluation Framework

Models were assessed through a hierarchy of measurement operators that separates near-equilibrium pointwise accuracy from process fidelity on dopant-migration, molecular-dynamics, NEB, and collective-sliding configurations. Static energy and force errors, molecular dynamics observables, fixed-path NEB scoring, self-relaxed NEB stability, interlayer sliding, and representation diagnostics were evaluated as distinct probes of the same candidate force fields. Detailed computational parameters are provided in the Supplementary Information.

2.3 Formal Benchmark Definition

The central object of this benchmark is not a single scalar RMSE, but a family of process-conditioned measurement operators applied to the same candidate force field. Let a configuration be denoted by

$$x = (\mathbf{R}, \mathbf{Z}, \mathbf{h}) \in \mathcal{X},$$

where $\mathbf{R}$ contains atomic coordinates, $\mathbf{Z}$ atomic identities, and $\mathbf{h}$ the simulation cell. A reference electronic-structure model defines an energy $E_{\mathrm{D}}(x)$ and force field $\mathbf{F}_{\mathrm{D}}(x) = -\nabla_{\mathbf{R}} E_{\mathrm{D}}(x)$. A candidate MLFF M defines $E_M(x)$ and $\mathbf{F}_M(x) = -\nabla_{\mathbf{R}} E_M(x)$.

Definition 1 (Equilibrium and process-conditioned evaluation distributions) The conventional equilibrium test distribution is

$$\mu_{\mathrm{eq}}(x) \propto \exp[-\beta E_{\mathrm{D}}(x)]$$

restricted in practice to relaxed structures, short AIMD trajectories, or random splits of near-equilibrium DFT configurations. Its standard pointwise error is

$$\mathcal{E}_{\mathrm{pt}}(M; \mu_{\mathrm{eq}}) = \left(\mathbb{E}_{x \sim \mu_{\mathrm{eq}}} \left[w_E |E_M(x) - E_{\mathrm{D}}(x)|^2 + w_F \|\mathbf{F}_M(x) - \mathbf{F}_{\mathrm{D}}(x)\|^2 \right] \right)^{1/2}.$$

For a physical process τ, such as dopant migration or interlayer sliding, the benchmark distribution is a process-conditioned distribution μ_τ supported on a structured set of configurations generated by a path operator. This distribution may place substantial probability mass on strained, high-energy, low-Boltzmann-weight configurations that are rare under μ_{eq} but control kinetics and mechanical stability.

Definition 1 makes explicit why equilibrium RMSE can be accurate and still incomplete: it measures the average pointwise error under μ_{eq}, whereas barrier prediction, diffusion, and sliding depend on E_M and ∇E_M over regions sampled by μ_τ. The two tests therefore answer different mathematical questions.

2.3.1 Pointwise Probes and Process Probes

A pointwise probe evaluates a model on isolated configurations,

$$\mathcal{P}_{\mathrm{pt}}(M;\mathcal{S}) = \{E_M(x), \mathbf{F}_M(x) : x \in \mathcal{S}\},$$

where $\mathcal{S}$ is a fixed set of DFT-labeled structures. This includes static energy/force RMSE and RDF-preserving near-equilibrium MD checks. A process probe evaluates a functional of a correlated configuration sequence,

$$\Gamma = (x_0, x_1, \ldots, x_N),$$

or of a trajectory distribution generated by M. Examples include a migration barrier, endpoint asymmetry, NEB convergence, diffusion coefficient, thermal conductivity, and sliding periodicity. The benchmark therefore measures both local correctness,

$$E_M(x_i) \approx E_{\mathrm{D}}(x_i), \qquad \mathbf{F}_M(x_i) \approx \mathbf{F}_{\mathrm{D}}(x_i),$$

and functional correctness,

$$\Phi_M(\Gamma) \approx \Phi_{\mathrm{D}}(\Gamma),$$

where Φ is a task-level observable rather than a pointwise label.

2.3.2 Migration-Path Selection

Let $\mathcal{A} = \{a_1, \ldots, a_K\}$ be metastable dopant sites obtained from relaxed DFT structures or thermally sampled AIMD motifs. A migration task is an ordered pair

$$\tau = (a_s, a_t, c),$$

where a_s and a_t are initial and final sites and c records the chemical and geometric context, such as in-gap motion, penetration into a quintuple layer, or interlayer sliding. The selected task set

$$\mathcal{T} = \mathcal{T}_{\mathrm{ID}} \cup \mathcal{T}_{\mathrm{OOD}} \cup \mathcal{T}_{\mathrm{coll}}$$

is designed to include: (i) interpolation-like paths whose local environments overlap with AIMD training data, (ii) extrapolation-like paths that pass through strained or high-energy transition states, and (iii) collective displacements where local environments change weakly but long-range registry or periodicity matters.

For each τ, a path is initialized by interpolation or chemically constrained image construction,

$$\Gamma_\tau^0 = (x_0, \ldots, x_N), \qquad x_0 = a_s, \quad x_N = a_t,$$

and then optimized by a reference or model-specific NEB operator. A path is retained as a benchmark probe only if its endpoints, chemical identity, cell convention, and image ordering are auditable from the raw inputs and outputs.

2.3.3 Fixed-Path and Self-Relaxed NEB Operators

The apparent discrepancy between a fixed-path barrier in Fig. 3 and a self-consistent MLFF-NEB barrier in Fig. S3 is resolved by defining two different operators.

Reference DFT path.

For a task τ, the DFT NEB operator returns a converged reference path

$$\Gamma^*_{\mathrm{D},\tau} = \mathrm{NEB}_{\mathrm{D}}(x_0, x_N) = (x^{\mathrm{D},*}_{0,\tau}, \ldots, x^{\mathrm{D},*}_{N,\tau}).$$

The DFT barrier is

$$\Delta E^{\ddagger}_{\mathrm{D}}(\tau) = \max_{0\le i\le N} E_{\mathrm{D}}(x^{\mathrm{D},*}_{i,\tau}) - E_{\mathrm{D}}(x^{\mathrm{D},*}_{0,\tau}).$$

Fixed-path scoring.

The fixed-path MLFF operator evaluates model energies on the same DFT-relaxed images:

$$\Delta E^{\ddagger}_{\mathrm{fixed}}(M,\tau) = \max_{0\le i\le N} E_M(x^{\mathrm{D},*}_{i,\tau}) - E_M(x^{\mathrm{D},*}_{0,\tau}).$$

It measures whether M assigns the correct potential-energy profile to an externally supplied DFT path. The model does not choose or relax the path.

Self-relaxed MLFF NEB.

The self-relaxed operator lets the MLFF define the forces that optimize the band:

$$\Gamma^*_{M,\tau} = \mathrm{NEB}_M(x_0, x_N) = (x^{M,*}_{0,\tau}, \ldots, x^{M,*}_{N,\tau}),$$

followed by

$$\Delta E^{\ddagger}_{\mathrm{self}}(M,\tau) = \max_{0\le i\le N} E_M(x^{M,*}_{i,\tau}) - E_M(x^{M,*}_{0,\tau}).$$

This operator measures optimization stability and qualitative path robustness on the MLFF's own PES. Because $\Gamma^*_{M,\tau}$ and $\Gamma^*_{\mathrm{D},\tau}$ need not be the same curve in configuration space, $\Delta E^{\ddagger}_{\mathrm{fixed}}$ and $\Delta E^{\ddagger}_{\mathrm{self}}$ are not expected to agree. A low self-relaxed MLFF barrier is physically validated only after the path is checked against DFT, for example by DFT single-point energies on $\Gamma^*_{M,\tau}$ or by restarting a DFT NEB from the MLFF path.

2.3.4 Error Decomposition

The fixed-path barrier error is

$$\epsilon_{\mathrm{fixed}}(M,\tau) = \Delta E^{\ddagger}_{\mathrm{fixed}}(M,\tau) - \Delta E^{\ddagger}_{\mathrm{D}}(\tau).$$

Define the local energy residual and the model/reference maximum-image indices as

$$\delta_M(x) = E_M(x) - E_{\mathrm{D}}(x),$$
$$i^*_M = \arg\max_i E_M(x^{\mathrm{D},*}_{i,\tau}), \qquad i^*_D = \arg\max_i E_{\mathrm{D}}(x^{\mathrm{D},*}_{i,\tau}).$$

The error can then be separated into endpoint and saddle-region terms:

$$\epsilon_{\text{fixed}} = \underbrace{\delta_M(x^{\mathrm{D},*}_{i^*_D}) - \delta_M(x^{\mathrm{D},*}_0)}_{\text{endpoint-to-saddle energy calibration}} + \underbrace{E_M(x^{\mathrm{D},*}_{i^*_M}) - E_M(x^{\mathrm{D},*}_{i^*_D})}_{\text{maximum-image localization on the fixed path}} .$$

Thus a model can fail by shifting the endpoint energy, miscalibrating the saddle energy, or moving the apparent maximum along the path. For a self-relaxed NEB, an additional geometric term appears because the model samples a different path:

$$\epsilon_{\text{self}\to\mathrm{D}}(M,\tau) = \left[\max_{x_i\in\Gamma^*_{M,\tau}} E_{\mathrm{D}}(x_i) - E_{\mathrm{D}}(x_0)\right] - \Delta E^{\ddagger}_{\mathrm{D}}(\tau),$$

which measures whether the MLFF-discovered path is also plausible on the DFT PES. In practice this term requires DFT single-point calculations or DFT NEB refinement on the MLFF-relaxed path.

2.3.5 Robustness and Generalization Claims

The benchmark supports claims at three levels, with different evidence requirements.

1. *Pointwise accuracy*: $\mathcal{E}_{\text{pt}}(M;\mu_{\text{eq}})$ is small, as measured by energy/force RMSE, RDF agreement, and short-time stability.
2. *Process fidelity*: $|\Phi_M(\tau) - \Phi_{\mathrm{D}}(\tau)|$ is small for a specified task τ, as measured by barriers, transport observables, sliding profiles, or DFT-path scoring.
3. *Robust generalization*: Φ_M remains accurate over a documented task family $\mathcal{T}$, with multi-path, multi-seed, OOD, and convergence checks.

The present work claims Level 2 evidence for selected Cr-doped Sb_2Te_3 tasks and proposes Level 3 as a benchmark design principle requiring broader validation across materials, migration chemistries, architectures, and random seeds. This distinction prevents a successful result on one path from being over-interpreted as universal transferability.

2.3.6 Benchmark Algorithm

Algorithm 1 Migration-centered MLFF benchmark

Define candidate models $\mathcal{M} = \{M_1, \ldots, M_L\}$ and reference method D.
Construct or audit equilibrium data $\mathcal{S}_{\mathrm{eq}}$ from DFT relaxations and AIMD.
Train or adapt models using documented training splits, seeds, and code hashes.
Measure pointwise errors $\mathcal{E}_{\mathrm{pt}}(M; \mathcal{S}_{\mathrm{eq}})$ and near-equilibrium observables.
Select migration and collective-displacement tasks $\mathcal{T} = \mathcal{T}_{\mathrm{ID}} \cup \mathcal{T}_{\mathrm{OOD}} \cup \mathcal{T}_{\mathrm{coll}}$.
for each task $\tau \in \mathcal{T}$ **do**
    Generate endpoint-verified initial image sequence Γ_τ^0.
    Compute or restart $\Gamma_{\mathrm{D},\tau}^* = \mathrm{NEB}_{\mathrm{D}}(x_0, x_N)$ until convergence.
    Score each M on the fixed DFT path to obtain $\Delta E_{\mathrm{fixed}}^{\ddagger}(M, \tau)$.
    Optionally run $\Gamma_{M,\tau}^* = \mathrm{NEB}_M(x_0, x_N)$ to assess self-relaxed stability.
    If $\Gamma_{M,\tau}^*$ differs from $\Gamma_{\mathrm{D},\tau}^*$, validate the MLFF path with DFT single points or a DFT restart.
    Record barrier errors, endpoint errors, maximum-force histories, convergence flags, RMSD/SOAP overlap, and checksums.
end for
Report model rankings with uncertainty over seeds, paths, and convergence status.
Interpret latent-space or SHAP diagnostics as supporting evidence, not as a replacement for task-level validation.

2.3.7 Applicability Assumptions and Acceptance Criteria

The framework is applicable when the process of interest can be represented by auditable endpoint states and a physically meaningful path or trajectory functional. It is especially appropriate for rare events, migration, sliding, stacking rearrangements, and other processes where a small set of high-leverage configurations controls the downstream observable. It is less complete when the dominant failure mode involves electronic excited states, bond-breaking chemistry absent from the reference functional, long-range electrostatics beyond the model architecture, or finite-size effects not represented in the chosen cell.

A benchmark task is accepted into the final manuscript only if the following checks are satisfied:

1. DFT endpoints and NEB images are reconstructable from raw inputs and outputs.
2. The DFT reference barrier is convergence-verified, or explicitly labeled as a current candidate value.
3. Fixed-path and self-relaxed NEB quantities are reported as distinct operators.
4. Model comparison includes seed sensitivity where training stochasticity can affect rankings.
5. Training/test leakage is checked with grouped splits or path-aware proximity metrics.
6. Path overlap is quantified with geometric RMSD and/or descriptor-space similarity.

7. Any MLFF-discovered low-barrier path is tested on the DFT PES before being interpreted physically.
8. Scripts, model identifiers, input files, convergence flags, and output checksums are preserved in the release manifest.

2.3.8 Physics and Data-Evidence Chain

The physical basis of the benchmark is transition-state theory: kinetic rates depend exponentially on activation free energies, so small barrier errors can lead to large rate errors,

$$k \propto \exp[-\Delta G^{\ddagger}/(k_{\mathrm{B}}T)].$$

Consequently, a model that has low average force RMSE near equilibrium can still be unreliable if it misrepresents the PES along rare-event pathways. The evidence chain used here is:

$$\begin{aligned}&\text{DFT/AIMD data} \rightarrow \text{trained MLFFs} \rightarrow \text{equilibrium validation}\\ &\rightarrow \text{path/process probes} \rightarrow \text{DFT ground-truth comparison}\\ &\rightarrow \text{representation diagnostics} \rightarrow \text{data-acquisition guidance.}\end{aligned}$$

This chain closes the benchmark loop: physical theory identifies the observables that matter, DFT supplies the reference PES, MLFFs are evaluated through measurement operators matched to downstream processes, and diagnostic analyses indicate which missing configurations should be acquired next.

2.4 SHAP Analysis

We employed a model-agnostic SHAP workflow to identify SOAP descriptors associated with MACE energy-error behavior. Local atomic environments were encoded using SOAP descriptors (5.0 Å cutoff, $n_{\max} = 4$, $l_{\max} = 4$), capturing species-resolved radial and angular interactions in Cr-doped Sb_2Te_3. For each MACE variant, gradient-boosted regressors (200 trees, learning rate 0.05, max depth 8) were trained to predict the absolute total-energy error on the audited 150-frame trajectory. TreeSHAP values were then computed for the fitted surrogate models, and surrogate fidelity was assessed with 5-fold cross-validation rather than in-sample R^2. We compared the top 50 influential SOAP features across models to identify shared versus model-specific sensitivities in the surrogate error model.

3 Results and Discussions

3.1 Equilibrium Properties: Foundation for Comparison

As a baseline, we first evaluated all four candidate models on the near-equilibrium DFT data used in conventional MLFF validation. In the notation of Sec. 2.3, this corresponds to estimating a pointwise error under μ_{eq}, the distribution of relaxed and thermally sampled configurations near local minima. All models satisfy typical energy and force RMSE thresholds (Table S1), and the corresponding parity plots show visually similar agreement with DFT across energies and forces (Fig. S1).

Low equilibrium RMSE establishes the equilibrium baseline. Migration, sliding, and rare-event kinetics are measured separately with fixed-path and self-relaxed pathway probes whose observables depend on the potential energy surface along physically motivated displacements.

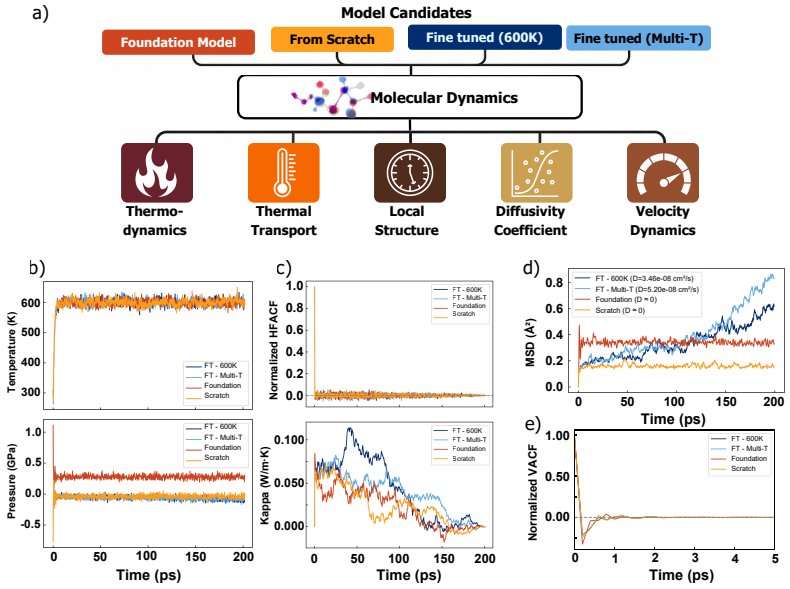


Fig. 2: **Comprehensive Benchmarking of MLFF Performance in Molecular Dynamics Simulations.** (a) A schematic of the unified evaluation protocol. Four candidate models (a zero-shot foundation model, a bespoke model trained from scratch, and two fine-tuned variants) were used to drive a 200 ps MD simulation. The resulting trajectories are then subjected to a uniform set of post-processing analyses to evaluate key physical properties. (b) Thermodynamic stability, demonstrated by the evolution of temperature and pressure over the simulation, remains near the target for all models. (c) Thermal transport properties, showing the Heat Flux Autocorrelation Function (HFACF) and its running integral to compute thermal conductivity (κ). (d) Mean-squared displacement (MSD), used to assess atomic mobility and calculate the diffusion coefficient. (e) Velocity autocorrelation function (VACF), which describes the underlying dynamics. Panels (b–e) use the same model mapping: dark blue, FT-600K; light blue, FT-MultiT; red, Foundation; orange, Scratch.

All MACE models (Foundation, Scratch, FT-600K, FT-Multi_T) successfully reproduce the equilibrium structure of Cr-doped Sb_2Te_3, with RDF analysis showing excellent agreement with the AIMD reference ("ground truth") across all atomic pair correlations (Fig. S2). This Figure includes a structural schematic of the local environment around Cr and full radial distribution functions (RDFs) for all models, confirming that each

training strategy preserves the characteristic nearest-neighbor and second-shell coordination features at 600 K. While all models maintain stable thermodynamic ensembles at 600 K (Fig. 2c), the zero-shot foundation model shows a persistent pressure offset, likely originating from its exposure to predominantly 0 K equilibrium structures rather than configurations extant at finite temperatures.

The models show similar RDF and VACF behavior but different transport estimates. The original FT-600K estimate used a shorter trajectory and a different sampling protocol; it is retained for provenance and excluded from model ranking. Under the current trajectories, the curves are best estimates rather than converged transport coefficients.

3.2 Non-Equilibrium Diffusion Pathways: The Critical Test of Generalization

The migration path benchmark evaluates a different measurement operator from the equilibrium RMSE. For each selected end-point pair $\tau = (a_s, a_t, c)$, the DFT NEB path defines a process-conditioned sequence of configurations $\Gamma^*_{\mathrm{D},\tau}$ that includes the metastable endpoints, intermediate distorted images, and the transition-state region. A candidate MLFF is then tested either by fixed-path scoring on this DFT geometry or by self-relaxed NEB optimization on its own PES, as defined in Sec. 2.3.3. These two operators answer different questions: whether the model assigns correct energies to the reference path, and whether the model can stably discover a plausible path without being given the DFT images.

3.2.1 Local Migration Events

For the fixed-path component of the benchmark, we used the DFT-relaxed NEB images as a common geometric reference and evaluated each MLFF by single-point energies on those images. This corresponds to $\Delta E^{\ddagger}_{\mathrm{fixed}}(M, \tau)$ in Sec. 2.3.3: the model is not allowed to move the path, so the resulting error isolates energetic fidelity along a DFT-supplied trajectory.

For the in-gap pathway, labeled 1–4 in the audited workflow, the provisional DFT barrier estimate is 0.34 eV. The archived QE NEB output has not yet passed the formal convergence gate; a changed converged value would replace the table entry and plotted DFT profile. The MACE Foundation fixed-path barrier is 1.02 eV, giving an audited fixed-path error of +0.69 eV relative to the provisional estimate. The Foundation profile is smooth but quantitatively miscalibrated on this reference path.

The Scratch profile overestimates the barrier, whereas the Foundation profile is smoother but remains quantitatively miscalibrated relative to the provisional DFT estimate. Fixed-path scoring and self-relaxed NEB are therefore reported as separate observables: the former tests energies on common DFT images, and the latter tests whether model forces produce a stable path.

We evaluated distinct Cr migration pathways (Fig. 3a) that sample different geometries. Pathway 1–4 is the controlled in-gap reference probe because its Cr displacement remains within the vdW gap. The broader 61-atom path family is

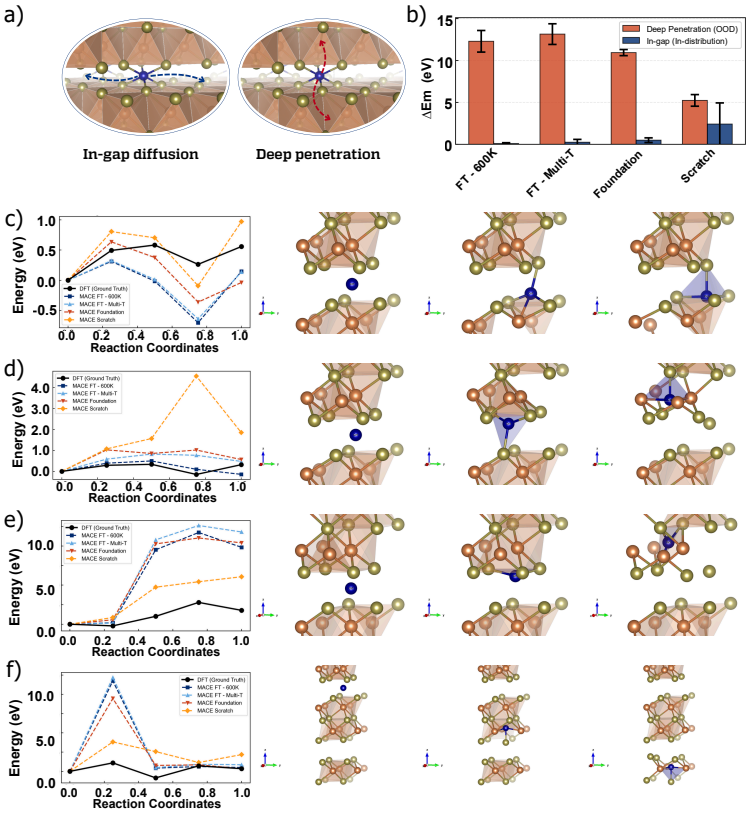


Fig. 3: **Comparative analysis of MLFF training strategies for predicting atomic migration barriers.** (a) Schematic illustration of the simulated Cr atom migration pathway between two stable sites within the Sb_2Te_3 bilayer. (b) Bar plot showing migration-energy prediction errors for deep-penetration and in-gap path classes. (c–f) MEP profiles for pathways 1–7, 1–4, 1–2, and 1–5, respectively. In panels (c–f), black circles denote DFT, dark-blue squares FT-600K, light-blue triangles FT-MultiT, red down-triangles Foundation, and orange diamonds Scratch.

classified separately in the Supplementary Information, including ambiguous and geometry-recheck cases.

In contrast, a deep-penetration event is defined by Cr entering the interior of a covalently bonded quintuple layer (QL), which distorts the surrounding lattice. These higher-distortion paths form a separate OOD class.

In-Gap Diffusion: A Test of Interpolative Accuracy. For the in-gap diffusion comparison, pathway 1–4 is the controlled probe (Fig. 3d); pathway 1–7 is shown separately in panel c and is not pooled with it. Under the grouped-split, multi-seed re-evaluation (seeds 123/234/345), the absolute fixed-path barrier error on pathway 1–4 relative to the provisional DFT estimate (0.34 eV) is 0.47 ± 0.12 eV for FT-600K fine-tuned

from the OMAT-FT checkpoint, 1.20 ± 0.27 eV for Scratch, and 0.68 ± 0.20 eV for Scratch-5%; a single-seed MACE-MPA-0 comparator gives 0.24 eV. FT-600K has the lowest reported error on this pathway. FT–MultiT was not retrained under the grouped protocol in this round.

The four representative foundation checkpoints (MP-0b3, MPA0, OMAT-0 medium, and MH-1) predicted pathway-1–4 barriers of 1.50, 1.79, 1.60, and 1.84 eV, respectively, compared with the provisional DFT value of 0.34 eV. All four overestimate the barrier by 1.17–1.50 eV; MPA0 is not the kinetic-accuracy winner in this completed comparison. Each checkpoint is reported individually because the foundation family does not provide training-seed replicates; full profiles and diagnostics are given in the Supplementary Information.

Deep Penetration: A Test of Extrapolative Robustness. The "deep penetration" pathway provides a much more stringent test of model robustness. This path involves significant lattice distortion as the Cr atom pushes its way through the covalently bonded quintuple layer (QL), creating atomic environments that are far from those seen in the training data (Fig. 3a).

The deep-penetration candidate paths sample larger lattice distortions and weaker overlap with the AIMD training distribution than pathway 1–4. They form an OOD robustness probe. Model ranking for this class requires the path classification and restarted QE references listed in the Supplementary Information audit.

The higher-distortion profiles vary across models, but the current audit does not identify a defensible Scratch ranking. The grouped comparison quantifies model-level seed variation for pathway 1–4; path-level variation remains unresolved for the higher-distortion class.

The Critical Role of Task-Specific Fine-Tuning

Fine-tuning improves process fidelity when adaptation data overlap with the pathway distribution. The FT-600K model, fine-tuned on data generated at the target temperature of the migration event, has the lowest reported error on the selected in-gap probe (Fig. 3b).

The 600 K fine-tuned model is closer to the in-gap migration probe than the broader multi-temperature model in this case study. The grouped-split SOAP/RMSD audit finds no in-gap NEB-image overlap with either the full training set or the Scratch-5% subset. The Scratch-5% control gives a pathway-1–4 error of 0.68 ± 0.20 eV, between FT-600K-OMATFT (0.47 ± 0.12 eV) and Scratch (1.20 ± 0.27 eV) on this pathway. FT–MultiT was not retrained under grouped protocol in this round, and the comparison is limited to pathway 1–4.

The 600 K and Multi-T results show that training-data selection changes process fidelity. Benchmark-guided acquisition can therefore target configurations that improve robustness on the measured pathways.

3.2.2 NEB Stability as an Additional Robustness Criterion

We also evaluated the separate self-relaxed NEB operator $\Delta E^{\ddagger}_{\mathrm{self}}(M, \tau)$, in which the MLFF supplies the forces used to optimize the band. The 0.41 eV barrier in Fig. S3

is the MACE Foundation model's self-consistent barrier on its own optimized path; it is not a DFT reference or the fixed-path +0.69 eV error. The Scratch, FT-600K, and FT–MultiT runs show increasing maximum image forces and terminate before a converged band is obtained. These runs are reported as self-relaxed NEB stability outcomes, not converged barriers.

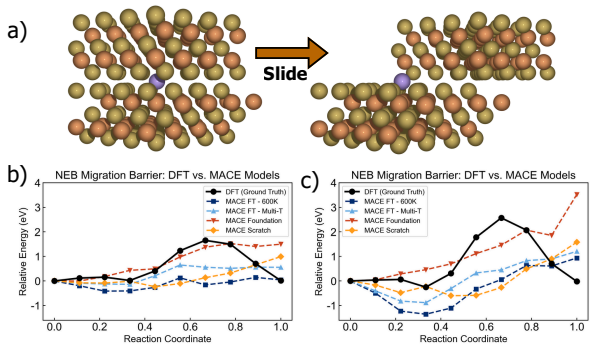


Fig. 4: Evaluation of MACE models on collective lattice displacement: interlayer sliding in Sb_2Te_3. (a) Schematic illustration of the bilayer sliding process in Sb_2Te_3, where the top layer (yellow/orange atoms) slides relative to the bottom layer along the crystallographic direction. The purple sphere indicates the position of a Cr dopant when present. (b) Energy barriers for interlayer sliding in pristine Sb_2Te_3 as calculated by DFT and the MACE models. (c) The corresponding barriers for Cr-doped Sb_2Te_3. Black circles denote DFT, dark-blue squares FT-600K, light-blue triangles FT-MultiT, red down-triangles Foundation, and orange diamonds Scratch in both panels.

3.3 Interlayer Sliding: Probing Robustness to Non-Local, Collective Displacements

Having tested the models' response to local, high-distortion events, we evaluated their robustness to a different class of out-of-distribution challenge: large-scale, collective lattice displacements. To this end, we simulated the shearing of one Sb_2Te_3 layer relative to another, a process governed by the subtle corrugations of the interlayer vdW potential energy surface (Fig. 4a). This task is particularly challenging for MLFFs as the local atomic environment of any given atom changes only minimally, while the global configuration undergoes a significant transformation that crosses periodic boundaries. The configurations along this sliding path were not present in the training data.

We first examined the case of pure, undoped Sb_2Te_3 (Fig. 4b). The Foundation model gives the closest barrier shape and magnitude among the models. Its final state remains higher in energy than the identical initial state, and Scratch shows the same endpoint drift.

The fine-tuned models (FT-600K and Multi-T) underestimate the barrier. Adaptation that improves local Cr chemistry does not preserve this collective sliding observable.

The same pattern appears in Cr-doped Sb_2Te_3 (Fig. 4c): all models miss barrier height and endpoint periodicity. The collective sliding probe therefore separates local bonding accuracy from long-range registry accuracy.

3.4 Representation Learning Analysis

To diagnose possible origins of the observed performance differences, we conducted a multi-faceted analysis of the learned representations. We projected high-dimensional atomic environment descriptors from each model's 600 K MD trajectory onto a two-dimensional space using two complementary visualization techniques: t-distributed stochastic neighbor-embedding (t-SNE) [81] to assess representational dissimilarity, and Potential of Heat-diffusion for Affinity-based Trajectory Embedding (PHATE) [82] to visualize trajectory geometry (Fig. 5).

The t-SNE projection (Fig. 5a) separates Foundation and Scratch, while the fine-tuned models occupy an intermediate region. In the audited analysis, the original 6191-dimensional Euclidean silhouette score is 0.33, the z-scored original-feature score is -0.02, and the projected baselines are 0.40 ± 0.03 for t-SNE and 0.47 for PHATE. The original-space and projected scores show representation differences across model families.

The PHATE projection (Fig. 5b) maps the continuous time evolution to a low-dimensional trajectory manifold whose geometry is compared with potential energy in Fig. 5c. Along PHATE 1, specialist models occupy similar regions; PHATE 2 separates Scratch from the fine-tuned models.

The Scratch model occupies an isolated region, whereas the fine-tuned models occupy intermediate regions. These representation differences accompany the observed diffusion and NEB differences; the causal relationship is not resolved by this analysis.

Latent-space separation accompanies the task-level differences, and we use it as a diagnostic alongside the barrier, transport, and stability benchmarks.

3.5 Dissecting Learned Representations with Feature-Level Interpretability

While t-SNE and PHATE suggest that distinct training strategies produce separable descriptor patterns, these projections do not establish the physical principles learned by each model. To add a feature-level diagnostic, we use SHapley Additive exPlanations (SHAP) [83] on surrogate models of MACE's absolute energy error. This analysis identifies SOAP descriptors to which the surrogate error predictors are most sensitive; it does not directly explain the internal message-passing mechanism of MACE.

Our methodology is designed to identify descriptors correlated with predictive error. For each MACE model, we trained a Gradient-Boosting Regressor to predict absolute total-energy error from 390 SOAP and 17 structural descriptors extracted from the X-FORCE 150-frame trajectory; frame identity is joined explicitly through `image_index`. [84] This approach interprets a simpler surrogate of the error landscape, not the internal message-passing mechanism of MACE itself. The audited 5-fold cross-validation R^2 values are 0.98, 0.88, and 0.97 for FT-600K, FT–Multi–T, and Scratch,

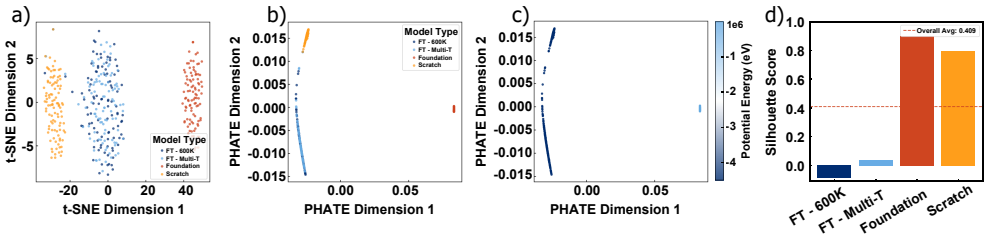


Fig. 5: **Representation diagnostics from trajectory embeddings and original-descriptor checks.** Projections of atomic environment descriptors from 600 K MD trajectories. **(a)** A *t*-SNE projection shows that Foundation (red) and Scratch (orange) models produce highly separated representations, while fine-tuned models (blue) are intermediate. **(b)** PHATE projection visualizes the system's continuous trajectory manifold. The separation between the scratch-model and fine-tuned-model projections along PHATE 2 is interpreted as a qualitative diagnostic rather than direct mechanistic proof. **(c)** The same PHATE embedding, colored by potential energy, compares projection geometry with model energy values. The foundation model's large energy offset indicates its nature as an uncalibrated prior. **(d)** Silhouette scores quantify separability in the original descriptor space and in low-dimensional embeddings. The projected values are used as visualization diagnostics, while the original-space values provide the primary quantitative check on representational dissimilarity.

respectively. We then applied TreeSHAP to these surrogate models, assigning feature-importance values to SOAP descriptors that are most correlated with prediction errors.

Our analysis shows a divergence in the surrogate feature-importance landscapes associated with each training strategy. A global overview of the top 50 influential descriptors across models (Fig. 6b) shows that the fine-tuned multi-temperature (FT-MultiT) surrogate concentrates importance more strongly, whereas the scratch-model surrogate distributes importance more diffusely. In the locked revised ranking, the largest FT-MultiT contribution is `Cr-Sb_n43_l0`; the leading Scratch contributions are dominated by Cr–Te descriptors. These are descriptive surrogate sensitivities, not evidence that the descriptors are causal or mechanistically internalized by MACE.

Figure 6a quantifies this disparity. The FT-MultiT profile is concentrated on Cr–Sb local-coordination descriptors, whereas the Scratch profile is broader. These feature patterns describe surrogate error predictors; they do not identify the physical mechanism of the NEB behavior.

4 Conclusions

We benchmarked specialist and foundation-based MLFFs for Cr-doped Sb_2Te_3 using the measurement-operator framework defined in Sec. 2.3. This formalism separates equilibrium point-wise accuracy from fidelity on dopant-migration and collective-sliding probes, and it distinguishes fixed-path energetic scoring from self-relaxed path optimization. Under this framework, diffusion trajectories and structured pathways

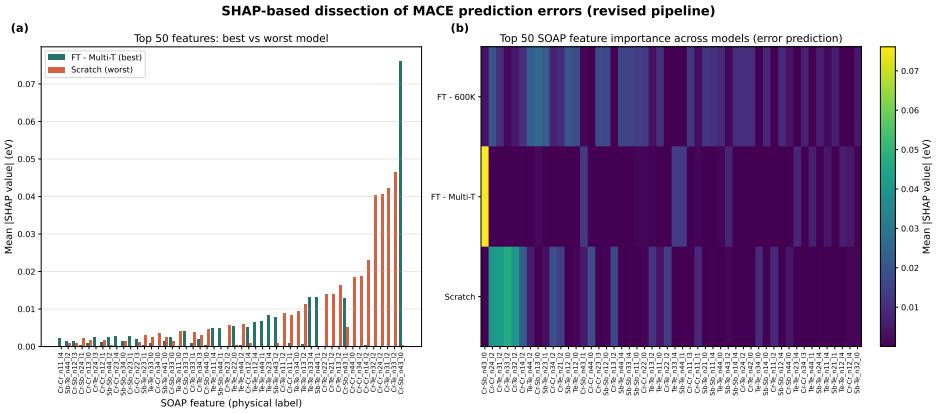


Fig. 6: **SHAP-based Dissection of Feature Importance in MACE Models.** **(a)** Mean absolute SHAP values for the top 50 SOAP features, comparing the best-performing fine-tuned model (FT-MultiT) and the worst-performing scratch-trained model. The corrected audit identifies `Cr-Sb_n43_l0` as the largest mean absolute SHAP feature in the FT-MultiT surrogate. **(b)** Heat map of surrogate feature importance for the top 50 SOAP features across all training strategies. The FT-MultiT surrogate concentrates importance on fewer features, while the scratch-model surrogate distributes importance more diffusely.

show that models with similar equilibrium metrics can diverge sharply under kinetically relevant distortions.

Our findings support several broader hypotheses. Robust benchmarking should span both in-distribution and out-of-distribution tasks, because accuracy near equilibrium does not guarantee kinetic fidelity. Fine-tuning can improve a selected process while reducing fidelity for other response regimes; this trade-off must be measured rather than assumed away. Latent-space analysis provides a qualitative diagnostic for why models with similar pointwise metrics may behave differently in downstream tasks. The higher-distortion path candidates remain outside the final numerical ranking until their geometry labels and restarted QE references are closed.

Together, these results indicate that adding "more data" alone is insufficient; pathway-relevant, non-equilibrium sampling can be decisive for meaningful improvement. Migration-inspired probes, coupled with latent-space diagnostics and auditable data provenance, provide a practical route to identify which configurations are needed and where models are likely to fail. In this case study, model selection is task-specific: the appropriate strategy must be validated against the particular non-equilibrium observables it is expected to predict. Claims of universally complementary model classes require validation across additional materials and architectures.

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- **Conflict of interest/Competing interests**
  The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Ethics approval and consent to participate**
  Not applicable. This study does not involve human participants or animal subjects.
- **Consent for publication**
  Not applicable.
- **Data availability**
  The supporting dataset, including training manifests, NEB path metadata, model-predicted barriers, and post-processing tables, will be released through MigrationBench on Hugging Face at https://huggingface.co/datasets/alinacao2000/MigrationBench. The manifest records the source trajectory, split assignment, random seed, model provenance, DFT/MLFF task type, convergence status, and checksum for each derived artifact. NEB paths are evaluation/reference data and were not merged into the fine-tuning datasets. The frame-level force audit covers 9,163 released training frames and found no unclassified or NEB-derived all-zero-force frame; the nine exact zero-force records are explicitly labelled isolated-atom E_0 references (release commit `f5c6d19`).
- **Materials availability**
  Not applicable. This work is entirely computational and does not involve the synthesis or distribution of physical materials.
- **Code availability**
  The scripts used for data processing, DFT/MD post-processing, and plotting are available on GitHub https://github.com/yicao-elina/MigrationBench.git. The Sec. 3.5/Fig. 6 SHAP chain is released as `scripts/shap_analysis.py`, `scripts/shap_feature_labels.py`, and `scripts/plot_shap_fig6.py`, using the compact three-model X-FORCE inputs in `data/raw_compact/revision1/shap/`. The source scope and MD5s are recorded in `docs/audit/shap_source_lineage.md`; the four-model, 2-concentration, and Dual-X ID/OOD/NEB workflows are not inputs to Fig. 6.
- **Author contribution**
  Y. C.: conceptualization, methodology, data curation, DFT and MD simulations, analysis, visualization, and manuscript-writing. **P. M.**: literature review, conceptualization, and manuscript- writing. **P. C.**: conceptualization, project administration, supervision, funding acquisition, and manuscript-writing.

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